SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	CSE	**Date:**	03/09/2026

Code of Conduct

*I **Purushothaman.V / 192511064** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Purushothaman.V

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Question 1: Answer

(a) PEAS Description

PEAS	Description
Performance Measure	+1 for every clean square at the end of each time step; $-p$ for every Left/Right movement
Environment	Two-room vacuum world containing squares A and B; initially both are dirty
Actuators	Suck, Left, Right, NoOp
Sensors	Sensor detects the agent's current location and whether the current square is Dirty or Clean

Interpretation:

The agent uses its location and dirt percepts to select an action that maximizes its performance over the six-step lifetime.

(b) Trace Agent X

Given:

- Initial position = A
- A = Dirty, B = Dirty
- $p = 1$
- Agent X: **If Dirty → Suck; otherwise → move to other square.**

Step-by-step table

Step	Location	Percept	Action	Clean squares at end	Move penalty	Step score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	Left	2	1	1
5	A	Clean	Right	2	1	1
6	B	Clean	Left	2	1	1

Total score

$$S_X = 1 + 0 + 2 + 1 + 1 + 1$$

$$\boxed{S_X = 6}$$

Interpretation:

Agent X successfully cleans both squares, but because it has no internal memory, it continues moving between already-clean squares. These unnecessary movements produce additional penalties.

(c) Trace Agent Y

Agent Y has internal state. Once it remembers that **both squares have been cleaned**, it performs **NoOp** instead of continuing to move.

Step	Location	Percept	Action	Clean squares at end	Move penalty	Step score
1	A	Dirty	Suck	1	0	1
2	A	Clean	Right	1	1	0
3	B	Dirty	Suck	2	0	2
4	B	Clean	NoOp	2	0	2
5	B	Clean	NoOp	2	0	2
6	B	Clean	NoOp	2	0	2

Total

$$S_Y = 1 + 0 + 2 + 2 + 2 + 2$$

$$\boxed{S_Y = 9}$$

Interpretation:

Agent Y achieves a higher score because it remembers that the environment is already clean and avoids unnecessary movements.

(d) Score as a Function of p

For Agent X:

Clean-square rewards over six steps:

$$1 + 1 + 2 + 2 + 2 + 2 = 10$$

Agent X makes **4 movements**.

Therefore,

$$\boxed{S_X(p) = 10 - 4p}$$

For Agent Y:

Clean-square rewards:

$$1 + 1 + 2 + 2 + 2 + 2 = 10$$

Agent Y makes only **1 movement**.

Therefore,

$$\boxed{S_Y(p) = 10 - p}$$

For Y to be strictly better:

$$10 - p > 10 - 4p$$

$$-p > -4p$$

$$3p > 0$$

$$\boxed{p > 0}$$

Therefore, whenever movement has a positive penalty, **Agent Y has a strictly higher score**.

At $p = 1$:

$$S_X = 6,\ S_Y = 9$$

(e) Which Agent Is Rational?

Agent Y is rational under the given performance measure for $p > 0$.

Rationality means selecting an action that maximizes the expected performance based on the available percept sequence and knowledge of the environment.

Agent X sees the same percepts as Agent Y, but X has no internal state. Therefore, after cleaning both squares, it cannot remember that the other square was already cleaned. It keeps moving between the squares and receives movement penalties.

Agent Y uses its **percept history/internal state** to recognize that both squares have already been cleaned and chooses **NoOp**.

Conclusion:

$$\boxed{S_Y > S_X \text{ for } p > 0}$$

Hence, according to the specified performance measure, **Agent Y behaves more rationally because it uses percept history to avoid unnecessary actions.**

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Question 2: Answer

(a) Classification

E1 — Autonomous Taxi

Property	Classification	Justification
Observability	**Partially observable**	Sensors cannot observe everything in traffic
Agents	**Multi-agent**	Other drivers and road users act independently
Determinism	**Stochastic**	Traffic events and driver actions are uncertain
Episodic/Sequential	**Sequential**	Current driving decisions affect future states
Static/Dynamic	**Dynamic**	Traffic and road conditions change continuously
Discrete/Continuous	**Continuous**	Position, speed and time vary continuously

Difficulty score

$$1 + 1 + 1 + 1 + 1 + 1 = \boxed{6}$$

E2 — Part-Picking Robot

Property	Classification	Justification
Observability	**Partially observable**	Camera perception may not provide complete information
Agents	**Single agent**	One robot performs the task
Determinism	**Stochastic**	Perception and picking outcomes may be uncertain
Episodic/Sequential	**Sequential**	Timing and previous actions affect subsequent picking
Static/Dynamic	**Dynamic**	Conveyor belt and parts are moving
Discrete/Continuous	**Continuous**	Position and movement vary continuously

Difficulty score

$$1 + 0 + 1 + 1 + 1 + 1 = \boxed{5}$$

E3 — Chess-playing Agent

Property	Classification	Justification
Observability	**Fully observable**	The complete board is visible
Agents	**Multi-agent**	Two players interact

Property	Classification	Justification
Determinism	**Deterministic**	A legal move has a predictable board result
Episodic/Sequential	**Sequential**	Each move affects future moves
Static/Dynamic	**Semi-dynamic**	Board does not change automatically, but the clock continues
Discrete/Continuous	**Discrete**	Moves and board states are discrete

Difficulty contribution:

$$0 + 1 + 0 + 1 + 0.5 + 0 = 2.5$$

$$\boxed{2.5}$$

(b) Difficulty Scores

Environment	Score
E1 — Autonomous Taxi	**6**
E2 — Part-Picking Robot	**5**
E3 — Chess	**2.5**

Therefore, based strictly on the **given scoring convention**, the scores are:

$$\boxed{E1 = 6,\ E2 = 5,\ E3 = 2.5}$$

(c) Agent Architecture

E1 — Autonomous Taxi

A suitable architecture is a **utility-based agent**.

Reason:

The taxi operates in a partially observable, stochastic, sequential, dynamic and continuous environment. It must consider multiple possible outcomes and select actions according to their expected usefulness, such as safety, travel time and route efficiency.

E3 — Chess

A suitable architecture is a **goal-based agent**.

Reason:

Chess is fully observable and sequential. The agent can evaluate possible moves with respect to a goal such as achieving checkmate. Its environment is also discrete and deterministic apart from the opponent's choice.

(d) Interpretation

The complexity of an agent program depends on the **properties of its environment**, not merely on the application domain.

For example, an autonomous taxi requires more complex decision-making because it faces partial observability, uncertainty, multiple agents, continuous changes and dynamic traffic.

In contrast, chess has a fully observable and discrete environment, although it remains sequential and multi-agent.

Therefore:

The more uncertain, dynamic, partially observable and continuous the environment is, the more internal state, prediction, planning or utility evaluation the agent may require.

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Question 3: Answer

(a) Correct Identification and Fooling Rates — 6 Marks

Formula

$$\text{Correct Identification Rate} = \frac{\text{Correctly identified}}{\text{Total interrogations}} \times 100$$

$$\text{Fooling Rate} = \frac{\text{Machine taken as human}}{\text{Total}} \times 100$$

Since every system has 50 interrogations:

ELIZA

Correctly identified:

$$47$$

$$\text{Correct rate} = \frac{47}{50} \times 100 = 94\%$$

Fooling:

$$50 - 47 = 3$$

$$\text{Fooling rate} = \frac{3}{50} \times 100 = \boxed{6\%}$$

Chatbot B

$$\text{Correct rate} = \frac{36}{50} \times 100 = \boxed{72\%}$$

$$\text{Fooling rate} = \frac{14}{50} \times 100 = \boxed{28\%}$$

System C

$$\text{Correct rate} = \frac{29}{50} \times 100 = \boxed{58\%}$$

$$\text{Fooling rate} = \frac{21}{50} \times 100 = \boxed{42\%}$$

Final table

System	Correct identification	Fooling rate
ELIZA	**94%**	**6%**
Chatbot B	**72%**	**28%**

System	Correct identification	Fooling rate
System C	**58%**	**42%**

(b) Turing's 70% Criterion

The question states that Turing's reference criterion is a maximum of **70% correct identification** after five minutes.

Therefore, a system satisfies the criterion if:

$$\text{Correct identification rate} \leq 70\%$$

System	Correct rate	Satisfies 70% criterion?	Distance from 50%
ELIZA	94%	**No**	$94 - 50 = 44$percentage points
Chatbot B	72%	**No**	$72 - 50 = 22$percentage points
System C	58%	**Yes**	$58 - 50 = 8$percentage points

Conclusion

$$\boxed{\text{Only System C satisfies the 70\% criterion.}}$$

System C's correct-identification rate is **8 percentage points above the 50% guessing level**.

(c) Fooling Rate Improvement

ELIZA → System C

ELIZA fooling rate:

$$6\%$$

System C fooling rate:

$$42\%$$

Improvement:

$$42 - 6 = \boxed{36 \text{ percentage points}}$$

Ratio:

$$\frac{42}{6} = \boxed{7}$$

So, System C's fooling rate is **7 times the ELIZA fooling rate**.

Chatbot B → System C

Chatbot B:

$$28\%$$

System C:

$$42\%$$

Improvement:

$$42 - 28 = \boxed{14 \text{ percentage points}}$$

Final answer

- **ELIZA → System C:** +36 percentage points; **7× ratio**
- **Chatbot B → System C:** +14 percentage points

(d) Interpretation

The Turing Test evaluates whether a machine can **act humanly** in a text conversation. A high fooling rate means that judges frequently mistake the machine for a human under the conditions of the test. However, this does **not prove that the system is thinking or acting rationally**.

For example, the **Chinese Room argument** distinguishes between producing appropriate-looking responses and actually understanding their meaning.

Therefore, System C's 42% fooling rate demonstrates performance in this particular Turing-test-style setting, but it does not by itself establish human-like understanding, consciousness, or rationality

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$\mathbf{U = 100 - T - 20 \cdot L}$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Question 4: Answer

(a) Utility of Each Route

R1 — Highway

Normal

$$T = 30,\ L = 0$$

$$U = 100 - 30 - 20(0)$$

$$\boxed{U = 70}$$

Disrupted

$$T = 70,\ L = 1$$

Since 70 > 55, the trip is late.

$$U = 100 - 70 - 20(1)$$

$$\boxed{U = 10}$$

R2 — City Road

Normal

$$T = 45,\ L = 0$$

$$U = 100 - 45$$

$$\boxed{U = 55}$$

Disrupted

$$T = 60,\ L = 1$$

Since 60 > 55, the trip is late.

$$U = 100 - 60 - 20$$

$$\boxed{U = 20}$$

R3 — Ring Road

No disruption:

$$T = 50,\ L = 0$$

$$U = 100 - 50$$

$$\boxed{U = 50}$$

Summary

Route	Normal utility	Disrupted utility
R1	**70**	**10 — Late**
R2	**55**	**20 — Late**
R3	**50**	**50**

(b) Expected Utility

Formula

$$EU = \sum P\,(\text{outcome})U(\text{outcome})$$

R1

Probability of normal:

$$1 - 0.30 = 0.70$$

Therefore:

$$EU(R1) = (0.70)(70) + (0.30)(10)$$

$$= 49 + 3$$

$$\boxed{EU(R1) = 52}$$

R2

Probability of normal:

$$1 - 0.10 = 0.90$$

$$EU(R2) = (0.90)(55) + (0.10)(20)$$

$$= 49.5 + 2$$

$$\boxed{EU(R2) = 51.5}$$

R3

No disruption:

$$EU(R3) = 1(50)$$

$$\boxed{EU(R3) = 50}$$

Comparison

$$EU(R1) = 52$$

$$EU(R2) = 51.5$$

$$EU(R3) = 50$$

Thus, according to the specified utility function and probabilities:

$$\boxed{\text{Choose R1 — Highway}}$$

This is a rational choice based on expected utility.

(c) New Percept: Highway Accident

The new radio alert increases R1's disruption probability from 0.30 to:

$$P(\text{disruption}) = 0.80$$

Therefore:

$$\boldsymbol{P(\text{normal}) = 1 - 0.80 = 0.20}$$

R1 expected utility becomes:

$$\boldsymbol{EU(R1) = (0.20)(70) + (0.80)(10)}$$

$$\boldsymbol{= 14 + 8}$$

$$\boxed{\boldsymbol{EU(R1) = 22}}$$

The other routes remain:

$$\boldsymbol{EU(R2) = 51.5}$$
$$\boldsymbol{EU(R3) = 50}$$

Therefore:

$$51.5 > 50 > 22$$

Choose R2 — City Road

Interpretation:

The physical roads have not changed, but the agent's percept sequence has changed because of the accident alert. Therefore, the estimated probability of disruption changes, which changes expected utility and consequently the rational decision.

(d) Actual Disruption and Rationality

In part (b), the agent selected R1 because:

$$EU(R1) = 52$$

Suppose the disruption actually occurs.

Then:

$$T = 70,\ L = 1$$

Therefore:

$$U = 100 - 70 - 20$$
$$U = 10$$

Was the agent irrational?

No

The agent was rational at the time of decision because it selected R1 based on the available information and probabilities.

A rational agent does not need to obtain the best possible outcome every time. It chooses the action expected to maximize performance based on its available percepts.

Rational vs Omniscient

Rational Agent	Omniscient Agent
Uses available precepts and knowledge	**Knows the actual future outcome**
Chooses according to expected performance	**Knows which outcome will actually occur**
Can experience an unfavourable outcome	**Can select based on perfect knowledge**

Therefore, receiving utility 10 after an unexpected accident does not make the original decision irrational.

(e) Interpretation

Rationality depends on:

1. Percept sequence — what information the agent has received.
2. Performance measure — what the agent is trying to maximize.
3. Available actions and knowledge — the choices and probabilities known to the agent.

In part (b), the highway had:

$$EU(R1) = 52$$

so, it was selected.

After the accident alert in part (c), the disruption probability increased to 80%, reducing:

$$EU(R1) = 22$$

Therefore, the agent changed its decision to R2.

Final interpretation:

The roads themselves did not change, but the agent's information about the environment changed. A rational agent updates its decision when its percept sequence changes.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	**15%**	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____